

SIZHUANG HE

Email: sizhuang.he@yale.edu | Homepage: <https://sizhuang.org> | GitHub: <https://github.com/SizhuangHe>

RESEARCH DIRECTIONS

LLM Agents and Harness Engineering: Tool use, workflow orchestration, and reliable scientific data curation

Agent Evaluation and Improvement: Verifiable feedback, failure analysis, and harness optimization

Generative Modeling: Discrete diffusion, flow matching, and generative modeling of permutations

Computational Biology: Protein language model interpretability and single-cell modeling

EDUCATION

Yale University

Ph.D. in Computer Science

- Advisor: Dr. David van Dijk

University of Michigan, Ann Arbor

Bachelor of Science in Honors Mathematics (Minor in Computer Science)

- Graduated with Highest Distinction
- GPA: 4.0 / 4.0

New Haven, CT

Aug. 2024 – Present

Ann Arbor, MI

Sep. 2019 – May 2023

WORK EXPERIENCE

Chan Zuckerberg Biohub, Inc.

AI Research Scientist Intern

Redwood City, CA

Jun. 2026 – Aug. 2026

- Developed an **LLM agent for interpreting sparse autoencoder (SAE) features in ESMC protein language models**, using activation evidence and contrastive protein pairs to produce general and context-specific annotations.
- Built discover/refine workflows with protein and residue inspection tools, structured annotation validation, provenance tracking, and batch execution with per-feature failure isolation.
- Designed a **controlled Gene Ontology molecular-function evaluation** with a temporal holdout, frozen question sets, and restricted evidence access; filtered answer-option shortcuts to assess the information provided by SAE annotations.

RESEARCH EXPERIENCE

Harmony Harness: LLM Agents for Biological Data Curation

- Co-developing **Harmony Harness** to orchestrate specialized agents for GEO DNA methylation data acquisition, phenotype harmonization, study metadata extraction, and methylation matrix processing.
- Engineering the execution harness with dependency-aware workflows, per-agent environments and tool permissions, bounded batch concurrency, and coordinated cancellation of failed runs.
- Implementing **artifact-level quality checks and decision provenance**: recording source evidence and transformation rules, then checking worked examples against independently reloaded outputs to detect unsupported or inconsistent mappings.
- Developing feedback-driven refinement with bounded retries and stopping when repeated issues persist; translating collaborator-reported failures into evidence-priority rules, field-level constraints, and regression checks.

Generative Models and LLMs

- Co-developed **Soft-Rank Diffusion**: reflected diffusion on continuous ranks, a latent reverse sampler with permutation projection, and contextualized Plackett–Luce denoisers. Improved long-sequence sorting and combinatorial optimization over prior permutation diffusion methods. *ICML 2026; co-first author.*
- Co-developed **CaDDi**, which unifies trajectory-conditioned discrete diffusion with causal language modeling. Adapted a **1.5B-parameter Qwen model** with CaDDi-AR and evaluated language-understanding tasks; demonstrated lower generative perplexity than evaluated discrete diffusion baselines. *NeurIPS 2025; co-first author.*
- Co-developed **CaLMFlow**, using a Volterra integral formulation and causal language models to learn continuous flows. Applied it to single-cell perturbation prediction with textual context and evaluated generalization to unseen perturbations.

PUBLICATIONS

* denotes equal contribution

- Learning Permutation Distributions via Reflected Diffusion on Ranks** [ICML 2026]
S. He*, Y. Zhang*, S. Zhang, D. van Dijk
- STRIDE: Post-Training LLMs to Reason and Refine Bio-Sequences via Edit Trajectories** [ICML 2026]
D. Zhang, S. Zhang, S. He, Y. Zhang, S. Rizvi, D. van Dijk
- Non-Markovian Discrete Diffusion with Causal Language Models** [NeurIPS 2025]
Y. Zhang*, S. He*, D. Levine, L. Zhao, D. Zhang, S. Rizvi, S. Zhang, E. Zappala, R. Ying, D. van Dijk
- TANTE: Time-Adaptive Operator Learning via Neural Taylor Expansion** [J. Comput. Phys., 2026]
Z. Wu, S. Wang, S. Zhang, S. He, M. Zhu, A. Jiao, L. Lu, D. van Dijk
- Intelligence at the Edge of Chaos** [ICLR 2025]
S. Zhang*, A. Patel*, S. Rizvi, N. Liu, S. He, A. Karbasi, E. Zappala, D. van Dijk
- COAST: Intelligent Time-Adaptive Neural Operators** [AI4MATH@ICML 2025]
Z. Wu, S. Zhang, S. He, S. Wang, M. Zhu, A. Jiao, L. Lu, D. van Dijk

MANUSCRIPTS UNDER REVIEW & PREPRINTS

* denotes equal contribution

- Reasoning through Study-Level Gene Set Signatures with TRACE** [In Review]
S. Rizvi, G. Marino, A. Patel, D. Levine, S. Zhang, S. He, D. van Dijk
- MORSE: Task-Oriented Multi-Agent System with Mixture of Role-Subtask Experts** [In Review]
P. Li, S. Zhang, Y. Zhang, S. He, D. van Dijk, R. Ying
- FLUX: Geometry-Aware Longitudinal Flow Matching with Mixture of Experts** [In Review]
J. Ortega Caro, Y. Zhang, H. M. Batchelor, S. He, J. Cardin, S. Saxena
- Scaling Large Language Models for Next-Generation Single-Cell Analysis** [In Review]
S. Rizvi*, D. Levine*, A. Patel*, S. Zhang*, E. Wang*, C. J. Perry, I. Vrkic, N. M. Constante, Z. Fu, S. He, et al.
- CaLMFlow: Volterra Flow Matching using Causal Language Models** [Preprint]
S. He*, D. Levine*, I. Vrkic, M. Bressana, D. Zhang, S. Rizvi, Y. Zhang, E. Zappala, D. van Dijk
- Operator Learning Meets Numerical Analysis: Improving Neural Networks through Iterative Methods** [Preprint]
E. Zappala, D. Levine, S. He, S. Rizvi, S. Lévy, D. van Dijk

HONORS & AWARDS

- **Fan Family Fellowship**, Yale University (2025)
- **Outstanding Achievement in Mathematics Award**, University of Michigan, Ann Arbor (2023)
- **James B. Angell Scholar**, University of Michigan, Ann Arbor (2023)
- **University Honors**, University of Michigan, Ann Arbor (2022, 2023)

TEACHING EXPERIENCE

- Teaching Fellow** Yale University
ENAS 1510 Multivariable Calculus for Engineers Fall 2026
- Course Assistant** University of Michigan, Ann Arbor
MATH 285 Honors Multivariable & Vector Calculus Fall 2023

SERVICE

- Journal Reviewer**
- ACM Computing Surveys 2026
 - Transactions on Machine Learning Research (TMLR) since 2025
- Conference Reviewer**
- AAAI Conference on Artificial Intelligence (AAAI) 2027
 - Conference on Neural Information Processing Systems (NeurIPS) 2026
 - International Conference on Machine Learning (ICML) 2026
 - International Conference on Learning Representations (ICLR) 2026
 - AI4MATH Workshop at ICML 2025